

The Impact of Ionic Liquid Loading in Three-Dimensional Carbon Nanotube Networks on the Separation of CO₂/CH₄ Fluid Mixtures: Insights from Molecular Simulations

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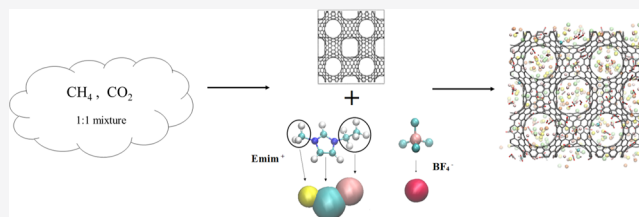
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ABSTRACT: The effect of the confinement of ionic liquids in carbon-based nanoporous materials on the separation of CO₂/CH₄ fluid mixtures has been systematically investigated in the present treatment by performing a series of Monte Carlo and molecular dynamics simulations. A three-dimensional carbon nanotube network and the Emim⁺-BF₄[−] ionic liquid have been used as prototype systems to study the separation of an equimolar CO₂/CH₄ fluid mixture at an ambient temperature and a wide pressure range. The effect of the ionic liquid concentration on the fluid uptake and separation selectivity has been systematically studied. The results obtained have revealed that the variation of the ionic liquid concentration is crucial in determining the fluid uptake and the thermodynamic selectivity. The molecular mechanisms affecting the translational self-diffusion and corresponding dynamics of the adsorbed fluids in the investigated nanoporous material, doped or not with the ionic liquid, have been analyzed in detail. The results obtained indicate that apart from tuning the thermodynamic selectivity, the ionic liquid concentration is a crucial factor in determining the translational dynamics of the adsorbed CO₂ and CH₄ molecules in the nanopores of the composite material. The translational dynamics of the adsorbed CO₂ and CH₄ molecules are also strongly influenced by their molecular shape, particularly during the time scales where the local cage effects affect their translational motions.



1. INTRODUCTION

CO₂ is a greenhouse gas and one of the main combustion products of fossil fuels.¹ It has been estimated that about 35 gigatonnes of CO₂ are produced per year by such processes, raising serious environmental concerns because CO₂ contributes significantly to global warming^{2,3} and ocean acidification.^{4,5} In view of the necessity of a gradual replacement of fossil fuels in the energy sector, the generation of low-carbon renewable energy sources is underway to help overcome these problems.^{6,7} Natural gas is considered as a near-term intermediate approach to reduce the reliance on coal-fired electricity and smoothen the transition to cleaner and renewable energy sources,⁸ which is the long-term objective in the framework of a global sustainable energy development plan. However, before it is used as a fuel by residential, commercial, and industrial consumers, natural gas needs to be processed in order to meet specifications mainly related to its heat content and flame temperature.^{9–15} To do so, several industrial processes have been developed to refine raw natural gas by removing impurities, undesired organic compounds, and other contaminants. CO₂ and methane are the most important constituents of natural gas, as well as biogas and landfill gas. The heat content of natural gas can be increased by reducing the amount of carbon dioxide. In this respect, the development of efficient methods to separate carbon dioxide

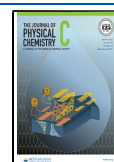
from natural gas mixtures has become an issue of crucial importance.

The necessity to achieve a high CO₂ selectivity and an efficient gravimetric uptake during the separation of CO₂/CH₄ fluid mixtures has motivated a significant effort to develop novel types of custom-designed porous materials^{16–38} and solvents,^{39–44} well suited to be used as task-specific sorbents. A key factor in the rational design of such sorbents is the simultaneous tuning of the thermodynamic and kinetic selectivity, as well as of the energy required for the sorbent regeneration. In this context, several crystal engineering and solvent design techniques have been suggested. Among the several suggested approaches to design tailored sorbents for CO₂ capture and separation, the incorporation of ionic liquids (ILs) within the micropores of solid porous materials is recently considered as an alternative method to control the CO₂ thermodynamic sorption selectivity.^{45–61} Due to their characteristic physicochemical properties,^{62–66} such as their vanishing vapor pressure, large

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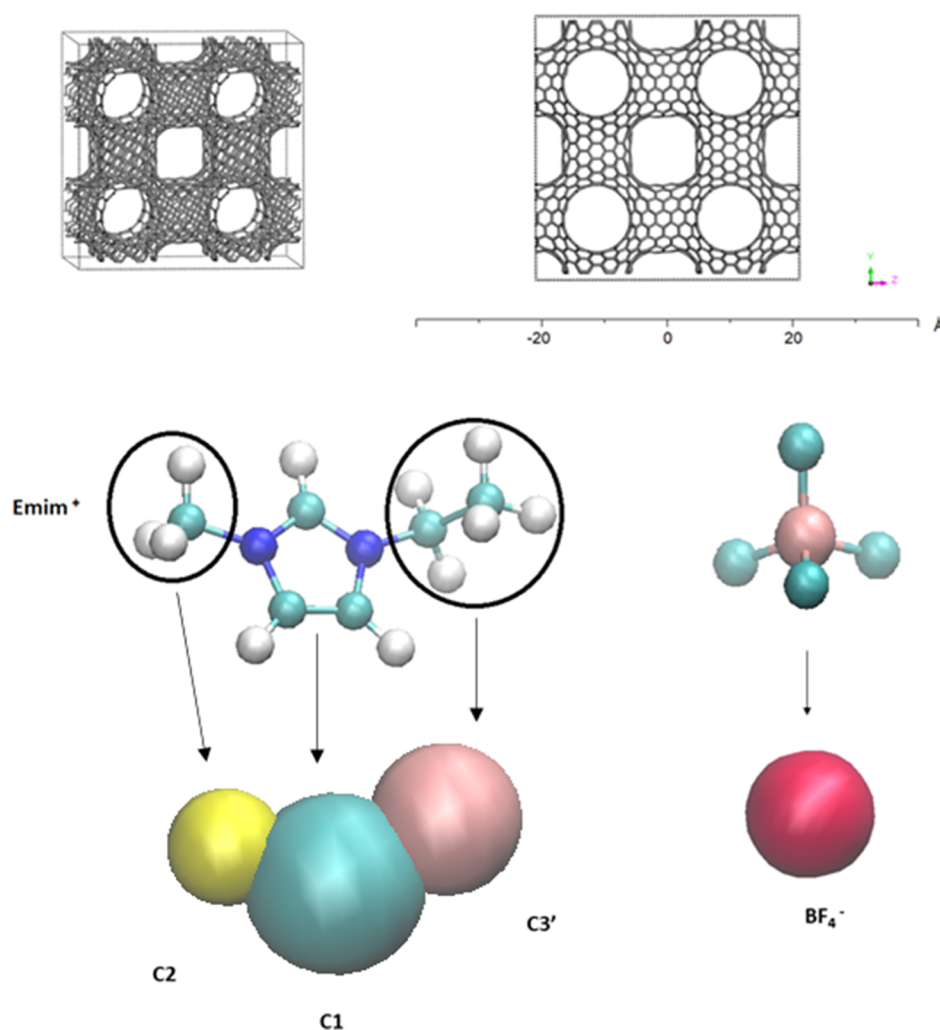


Figure 1. PNN cubic supercell, depicted together with a schematic representation of the coarse-grained models for the Emim^+ cation and the BF_4^- anion used in the simulations in the present work.

liquidus range, high thermal stability, relative nonflammability, and recyclability, ILs exhibit unique advantages over other volatile organic solvents. The availability of a large number of cations and anions to form ILs also presents tremendous opportunities for finding optimal cation/anion pairs and IL mixtures for the design of IL targeted toward task-specific applications, with promising perspectives in the production of clean and self-sustained energy. For these reasons, they are currently used in a wide range of applications, including carbon capture, storage, and utilization (CCSU). The major drawback of ILs in CCSU applications arises from their high viscosity, which further increases upon CO_2 capture. This issue can be overcome by designing composite materials, where the IL cations and anions impregnate the pores of a solid porous support. Although several types of composite materials,^{45–61} where ILs are supported on porous molecular sieves, have been reported in the literature as efficient for CCSU applications, a systematic rational design of such hybrid materials is highly important. However, the progress of such a rational design and screening process largely depends on the deeper understanding of the molecular-scale mechanisms controlling the CO_2 thermodynamic selectivity.

Owing to their exceptional physicochemical characteristics, stability, and light framework, carbon-based nanomaterials⁶⁷ are

considered as promising media for gas storage and separation applications.^{13,15,31,38,58,68–77} In this context, the present study aims to shed some light upon the role of the IL concentration in model hybrid composite materials, consisting of a prototype IL and a prototype model of three-dimensional carbon nanotube networks, on the separation of CO_2/CH_4 fluid mixtures. The modification of the CO_2 uptake capacity and separation selectivity in a model three-dimensional, interconnected single-walled carbon nanotube network^{38,73–76} when inserting cation–anion pairs of the prototype ethyl-methyl-imidazolium (Emim^+)-(BF_4^-) IL into the pores of the material has been systematically studied in the present treatment using a combination of Monte Carlo (MC) and molecular dynamics (MD) simulations. The (Emim^+)-(BF_4^-) IL, according to the literature, is one of the typical imidazolium-based ILs exhibiting high CO_2 solubility⁷⁸ and has been used in IL membranes,⁷⁹ as well as in IL-coated porous graphene materials⁸⁰ for the separation of fluid mixtures containing CO_2 . For these reasons, it has been selected as a model IL system in the present study. Particular attention has been paid on the effect of the IL concentration not only on the thermodynamic separation selectivity but also on the CO_2 and CH_4 uptake, as well as on the kinetic selectivity. Note that previous experimental^{81–84} and computational structural prediction studies⁸⁵ have reported that

Table 1. Parameters of the Coarse-Grained Potential Models of Emim⁺ and BF₄[−]

atom type	<i>x</i> (Å)	<i>y</i> (Å)	<i>z</i> (Å)	<i>m</i> (g/mol)	<i>q</i> (e)	<i>σ</i> (Å)	<i>ε</i> (kcal/mol)
Emim ⁺							
C1	0.000	−0.527	1.365	67.07	0.3591	4.38	0.61185
C2	0.000	1.641	2.987	15.04	0.1888	3.41	0.086042
C3'	0.000	−0.737	−1.653	29.07	0.2321	4.38	0.29637
BF ₄ [−]							
BF ₄ [−]	0.000	0.000	0.000	86.81	−0.78	4.51	0.77438

the development of such kind three-dimensional carbon nanostructures is possible. Therefore, our investigation might serve as a springboard toward the rational design of IL carbon-based hybrid composite nanomaterials suitable as media for the separation of CO₂/CH₄ fluid mixtures.

2. COMPUTATIONAL METHODS

A combination of force-field-based Grand Canonical MC (GCMC) and MD simulations was employed in order to explore the effect of the IL concentration in a hybrid composite material on the adsorptive separation of a reference equimolar CO₂/CH₄ bulk fluid mixture. The composite material consisted of a carbon-based porous nanotube network (PNN)^{38,73–76,86} and cations and anions of the Emim⁺-BF₄[−] IL confined into the nanopores. The PNN is composed of (8,8) single-walled carbon nanotubes, with a nanotube diameter around 11 Å, which are connected through junctions, forming a three-dimensional cubic carbon nanotube network and a corresponding three-dimensional pore structure. The PNN supercell used in our simulations comprises one cubic unit cell with a 41 Å dimension and is presented in Figure 1. Two different IL concentrations, with 20 and 60 cation–anion pairs inserted into the pores of PNN, have been investigated in the present treatment. These IL concentrations correspond to 9.9 and 29.7% of the mass of the PNN supercell, respectively. GCMC simulations were also performed to study the adsorption of the equimolar CO₂/CH₄ fluid mixture in PNN which is not doped with an IL, in order to have a clear picture of the IL loading effect on the adsorption and separation of the mixture. All the GCMC simulations were performed with the RASPA code⁸⁷ at 303.15 K and in the pressure range 0.1–20 bar. In these simulations, the fixed number of IL cations and anions, corresponding to the previously mentioned IL concentrations, was introduced by using a Canonical (NVT) MC scheme, whereas CO₂ and CH₄ molecules were inserted using the GCMC scheme. The fugacity of CO₂ and CH₄ at the investigated thermodynamic conditions was calculated using the Peng–Robinson equation of state.⁸⁸ For each simulation point, 2 × 10⁵ MC cycles were used for the equilibration, as well as for the production runs. Several types of trial molecular moves were employed in each MC cycle: translation, rotation, creation and deletion, and exchange of molecular identity. The rigid EPM2⁸⁹ and OPLS-AA⁹⁰ force fields were employed for CO₂ and CH₄ molecules, respectively. These models provide very accurate descriptions of the structural, thermodynamic, transport, and dynamic properties of CO₂ and CH₄, as well as of their mixtures, in a wide range of thermodynamic conditions.^{91–93} The coarse-grained potential models developed by Merlet *et al.*⁹⁴ were employed for the Emim⁺ and BF₄[−] ions of the IL. The selection of these models has been carried out based upon the fact that their simplicity, which makes them much less demanding in terms of required computational resources, is also combined with an accuracy in the prediction of the static and dynamic properties of the bulk IL

in a wide temperature range. A schematic representation of these coarse-grained potential models, as well as their parameters, is presented in Figure 1 and Table 1, respectively. The C–C parameters for the Lennard–Jones interaction between the carbon atoms of the investigated nanostructures and the atoms of the CO₂ and CH₄ molecules, as well as of the IL ions, were also adopted by previous studies in the literature,^{95–97} and the Lorentz–Berthelot combining rules were employed for the cross LJ interactions between the different atoms of the adsorbate/adsorbent pairs. A cut-off value of 12 Å was employed for short-range interactions, while the Ewald summation method was considered for the long-range electrostatic interactions. The geometry of the PNN supercell has been maintained rigid in the simulations.

After determining the amount of adsorbed CO₂ and CH₄ molecules in the PNN and the hybrid IL-PNN materials, corresponding to the equimolar bulk fluid mixture at 1 bar and 303.15 K, MD simulations were performed in the canonical ensemble in order to investigate the transport and dynamic properties of the confined CO₂–CH₄-IL mixtures. The initial configurations for the MD simulations were prepared by employing NVT-MC simulations, performing 4 × 10⁵ MC cycles, using the calculated average uptake of CO₂ and CH₄ molecules in each one of the investigated IL-PNN materials. The simulated systems in the PNN nanopores contained the following components: (i) 90 CO₂ and 22 CH₄ molecules; (ii) 103 CO₂ molecules, 20 CH₄ molecules, 20 Emim⁺ cations, and 20 BF₄[−] anions; and (iii) 120 CO₂ molecules, 15 CH₄ molecules, 60 Emim⁺ cations, and 60 BF₄[−] anions. The systems were then equilibrated for a period of 40 ns, and the system properties were evaluated in subsequent 200 ns production runs. An additional 200 ns MD simulation of each one of the two systems containing IL pairs, starting from a different initial configuration produced by a different NVT-MC simulation in each case, was also performed to ensure that the dynamics of the confined fluids are not affected by the initial configuration. The results obtained by the two independent simulations at the two investigated IL concentrations further confirm that the statistical sampling of the properties of the simulated systems is sufficient. The same force fields and methods to treat the short- and long-range interactions with the ones used in the GCMC simulations were also employed in the case of the MD simulations. The equations of motion were integrated using a leapfrog-type Verlet algorithm⁹⁸ and the integration time step was set to 1 fs. The intramolecular geometry of CO₂ and CH₄ molecules was constrained using the quaternion formalism.⁹⁸ A Nose–Hoover thermostat⁹⁹ with a temperature relaxation time of 0.5 ps was used to constrain the temperature during the simulations. All MD simulations were performed with the DL_POLY simulation code.¹⁰⁰

3. RESULTS AND DISCUSSION

3.1. Thermodynamic–Structural Properties. The calculated gravimetric co-adsorption isotherms of CO₂ and CH₄, taking as reference the mass of the total hybrid material (PNN + IL), are presented in Figure 2 for the different IL concentrations, ranging from 0 to 60 IL pairs. From this figure, it can be clearly observed that at pressures lower than 1 bar, the IL addition contributes to the increase of the CO₂ gravimetric uptake and the decrease of the CH₄ one. The increase of the CO₂ adsorption below 1 bar can be attributed to the enhanced affinity of CO₂ with the anion of the IL pairs. However, at higher pressures, the gravimetric uptakes of CO₂ and CH₄ are both reduced when the IL concentration increases. This effect is related to the decrease of the available free pore volume due to the IL addition, which at higher pressures results in a lower uptake. The saturation capacity, which is approached at around the highest limit of the investigated pressure range (~20 bar), is also decreased due to the decrease of the free pore volume when adding more IL ion pairs.

Apart from the gravimetric co-adsorption isotherms, thermodynamic adsorption selectivity $S_{(\text{CO}_2/\text{CH}_4)}$ was also calculated

$$S_{(\text{CO}_2/\text{CH}_4)} = \left(\frac{x_{\text{CO}_2}}{x_{\text{CH}_4}} \right) \left(\frac{y_{\text{CH}_4}}{y_{\text{CO}_2}} \right) \quad (1)$$

where x_i and y_i are the molar fractions of each component i ($i = \text{CO}_2, \text{CH}_4$) in the adsorbed and bulk phases, respectively. The calculated thermodynamic adsorption selectivity $S_{(\text{CO}_2/\text{CH}_4)}$ for the CO₂–CH₄ bulk equimolar fluid mixture as a function of pressure at 303.15 K is presented in Figure 2 for the investigated IL concentrations. Interestingly, it can be seen that the thermodynamic adsorption selectivity is enhanced with increasing IL concentration for the whole investigated pressure range and it becomes higher as the pressure increases. The increase of the selectivity $S_{(\text{CO}_2/\text{CH}_4)}$ at high pressures when the IL concentration increases can be explained in terms of the significant decrease of the gravimetric methane uptake at higher IL concentrations, as it can also be seen in Figure 2. This decrease is more significant than the corresponding decrease of the CO₂ uptake, leading thus to a selectivity increase at higher IL concentrations.

To provide a quantitative estimate of the performance of the composite PNN-IL materials on the separation of the fluid CO₂–CH₄ mixtures in comparison with the separation performance of PNN, we introduced a performance factor p defined as

$$p = \left(\frac{S_{(\text{CO}_2/\text{CH}_4)}^{\text{PNN+IL}}}{S_{(\text{CO}_2/\text{CH}_4)}^{\text{PNN}}} \right) \left(\frac{\text{CO}_2 \text{ uptake}^{\text{PNN+IL}}}{\text{CO}_2 \text{ uptake}^{\text{PNN}}} \right) \quad (2)$$

This performance factor relates the modification of the selectivity to the corresponding changes in the uptake of CO₂ induced by the addition of the IL ion pairs. The pressure dependence of this performance factor is presented in Figure 3, where it can clearly be seen that p increases significantly upon the IL addition at the low-pressure range. However, as the pressure increases, this factor decreases and this decrease is more pronounced for the higher IL concentration. Taking into account the behavior of p with the pressure, as well as the absolute values of the gravimetric uptake of CO₂ and the

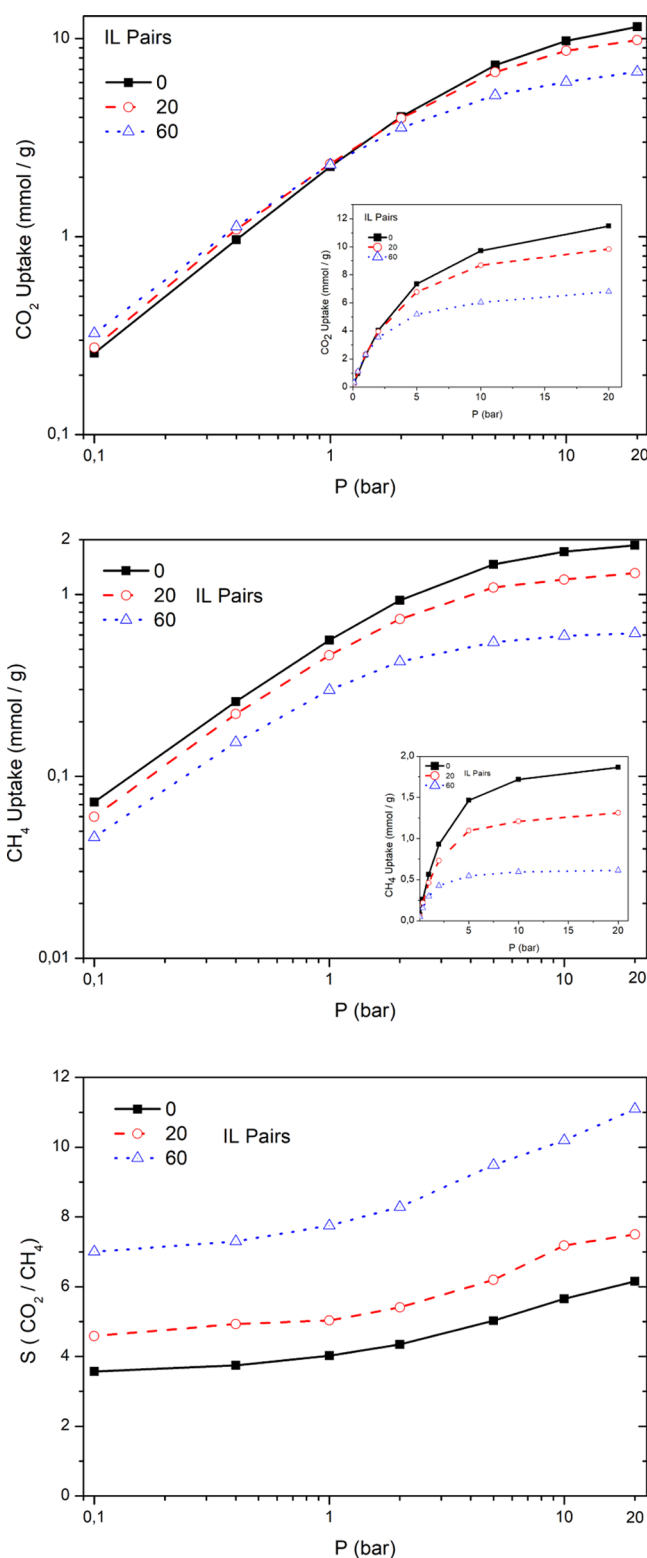


Figure 2. Calculated gravimetric co-adsorption isotherm logarithmic plots of CO₂ and CH₄, as well as the thermodynamic adsorption selectivity $S_{(\text{CO}_2/\text{CH}_4)}$, corresponding to the investigated PNN and PNN-IL composite materials. The linear plots are presented in the inset pictures.

thermodynamic adsorption selectivity, we may conclude that the pressure range around 1 bar seems to simultaneously fulfil the criteria for a quite satisfactory gravimetric uptake, selectivity, and

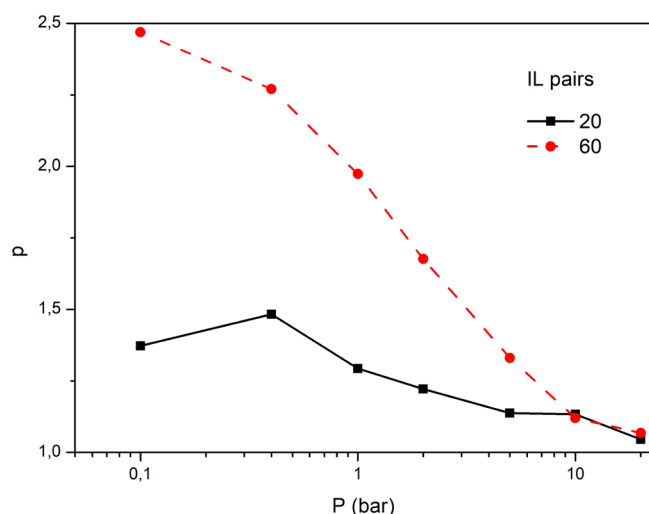


Figure 3. Calculated pressure dependence of the performance factor p , presented for the investigated IL concentrations.

performance factor. For this reason, we decided to further investigate the effect of the IL addition upon the structure and dynamics of the confined mixtures at 1 bar, by performing MD simulations of the adsorbed CO_2 – CH_4 mixtures at each one of the investigated IL concentrations.

By analyzing the MD trajectories, the density profiles $\rho(z)$ of the adsorbed CO_2 and CH_4 molecules along the z -axis of the PNN supercell, displayed in Figure 1, were calculated and are presented in Figure 4. From this figure, it can clearly be seen that the IL addition causes a significant decrease in the CH_4 uptake in the nanopores of the composite material. The CO_2 average density, expressed in molecules/ $\text{\AA}^3$, increases when the ionic liquid is added. The different CO_2 packing effects in nanopores of the composite material, induced by the IL addition, are reflected on the density profiles. To further investigate these packing effects, the center of mass–center of mass (COM–COM) pair radial distribution functions (rdf) between the confined CO_2 and CH_4 molecules and the IL cations and the anions were calculated and are presented in Figures 5 and 6. The results obtained clearly indicate the existence of stronger interactions between the CO_2 molecules and the IL ions, particularly with the BF_4^- anions. The stronger CO_2 –anion interactions are reflected on the high-intensity first peaks of the calculated CO_2 – BF_4^- rdf, presented in Figure 6, which are located at 3.85 $\text{\AA}$. On the other hand, the intensity of the first peaks of the calculated CO_2 – Emim^+ rdf, located at longer distances around 5 $\text{\AA}$, is lower. This finding signifies that the CO_2 –cation interactions are relatively weaker. The first minima of the calculated CO_2 – BF_4^- and CO_2 – Emim^+ rdf are located around 6.15 and 7.35 $\text{\AA}$, respectively, further indicating the previous finding. The IL concentration effect upon the first maxima and minima positions of these rdfs has also been found to be almost negligible. However, the intensity of these peaks slightly decreases upon increasing the IL concentration. The very low intensities of the calculated CH_4 – BF_4^- and CH_4 – Emim^+ rdf at short distances, presented in Figure 6, signify that the CH_4 –IL interactions are significantly weaker. Finally, the formation of IL anion–cation pairs exhibiting strong coulombic interactions is reflected on the high intensity peaks of the corresponding rdf around 4.25 $\text{\AA}$ and is also presented in Figure 6. The much higher intensity of the anion–cation rdf at the lowest IL concentration arises from the formation of anion–cation pairs at

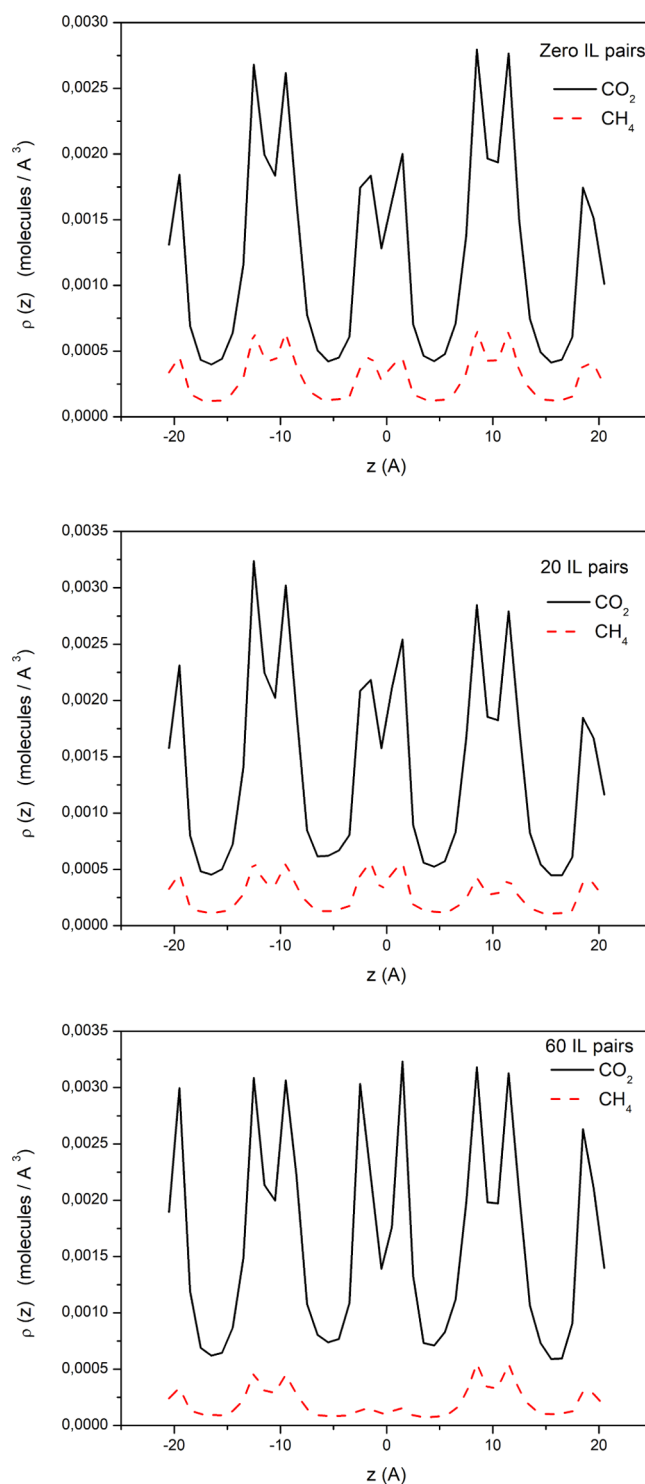


Figure 4. Calculated density profiles of the adsorbed CO_2 and CH_4 molecules at 1 bar, presented for the investigated PNN and PNN-IL composite materials.

this distance range. The formation of these ion pairs is observed at the investigated IL concentrations, due to the strong coulombic interactions between the anions and cations. Therefore, at the low IL concentration, the local cation density around an anion is much more enhanced in comparison with its overall value in the nanomaterial, resulting in a much higher intensity peak of the rdf.

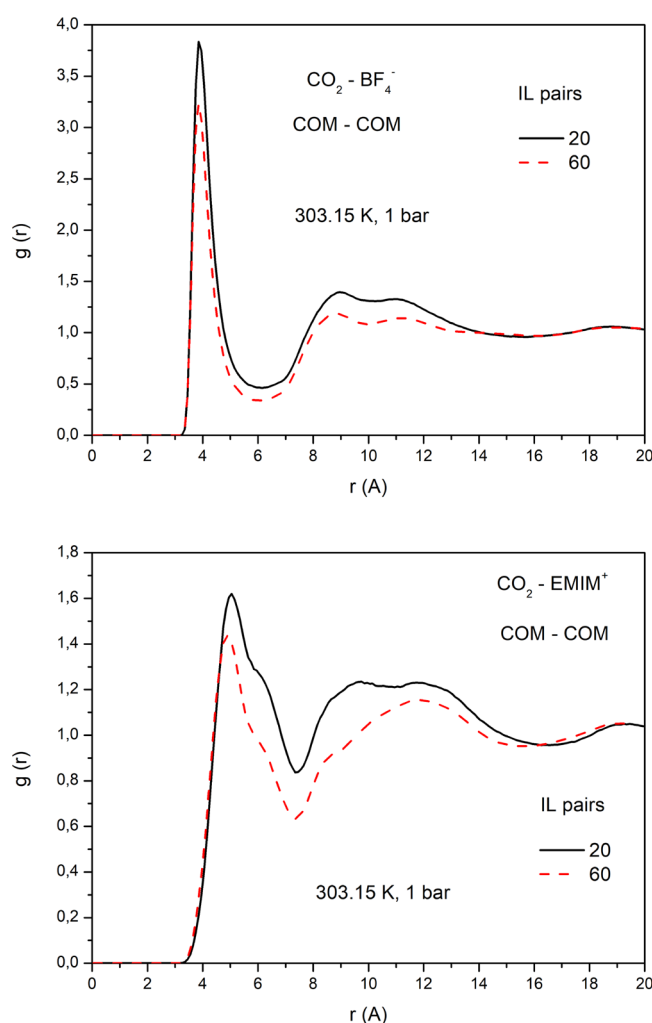


Figure 5. Calculated pair rdf between the adsorbed CO₂ molecules and the IL cations and the anions, presented for the investigated IL concentrations at 1 bar.

3.2. Transport–Dynamic Properties. The short-time translational dynamics of the confined CO₂ and CH₄ molecules, as well as of the confined IL anions and cations, were investigated in terms of their normalized COM velocity time correlation functions (tcf) and their corresponding spectral densities

$$C_v^i(t) = \frac{\langle \vec{v}_i(0) \cdot \vec{v}_i(t) \rangle}{\langle \vec{v}_i(0)^2 \rangle} \quad (3)$$

$i = \text{CO}_2, \text{CH}_4, \text{Emim}^+, \text{BF}_4^-$

$$S_v^i(\omega) = \int_0^\infty \cos(\omega \cdot t) \cdot C_v^i(t) \cdot dt \quad (4)$$

Low-frequency motions, related to cage effects in condensed fluid systems, are usually reflected on the shape of these tcfs and their spectral densities and provide useful information about the local structural relaxation phenomena in these systems.⁸⁶ The calculated COM velocity tcfs of all the adsorbed species, as well as their spectral densities, are presented in Figures 7 and 8. The much stronger cage effects in the case of the IL anions and cations are reflected on the time decay of the velocity tcfs, with the existence of a negative minimum around 0.3 ps, signifying that the translational motions of the IL ions are significantly

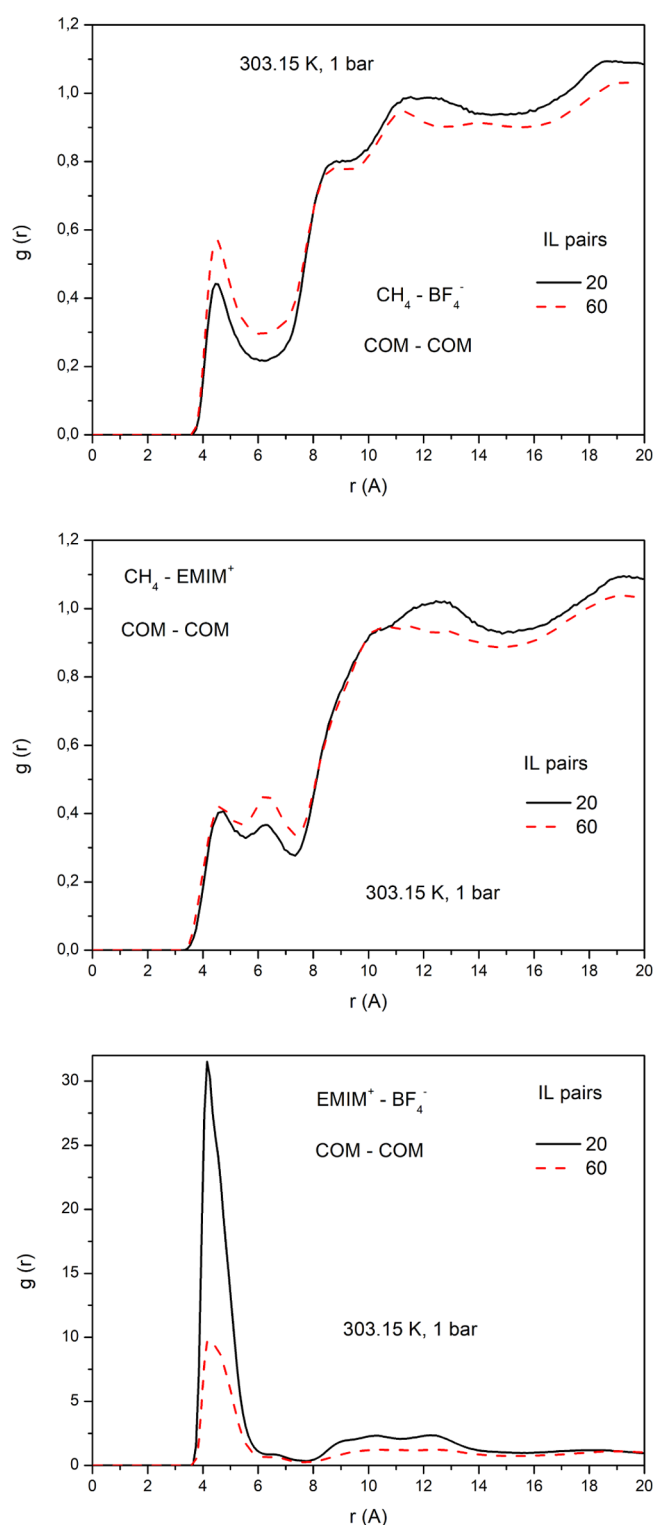


Figure 6. Calculated pair rdf between the adsorbed CH₄ molecules and the IL cations and the anions, as well as between the confined IL cations and anions, presented for the investigated IL concentrations at 1 bar.

hindered in comparison with the CO₂ and CH₄ molecules. The very similar shape of the tcfs for the Emim⁺ and the BF₄[−] ions signifies that their translational dynamics are strongly affected by the ion-pair formation. The IL concentration effects are mainly reflected on the time scale required for these tcfs to reach the zero value. This time scale becomes shorter as the IL concentration increases, due to the faster collisional effects

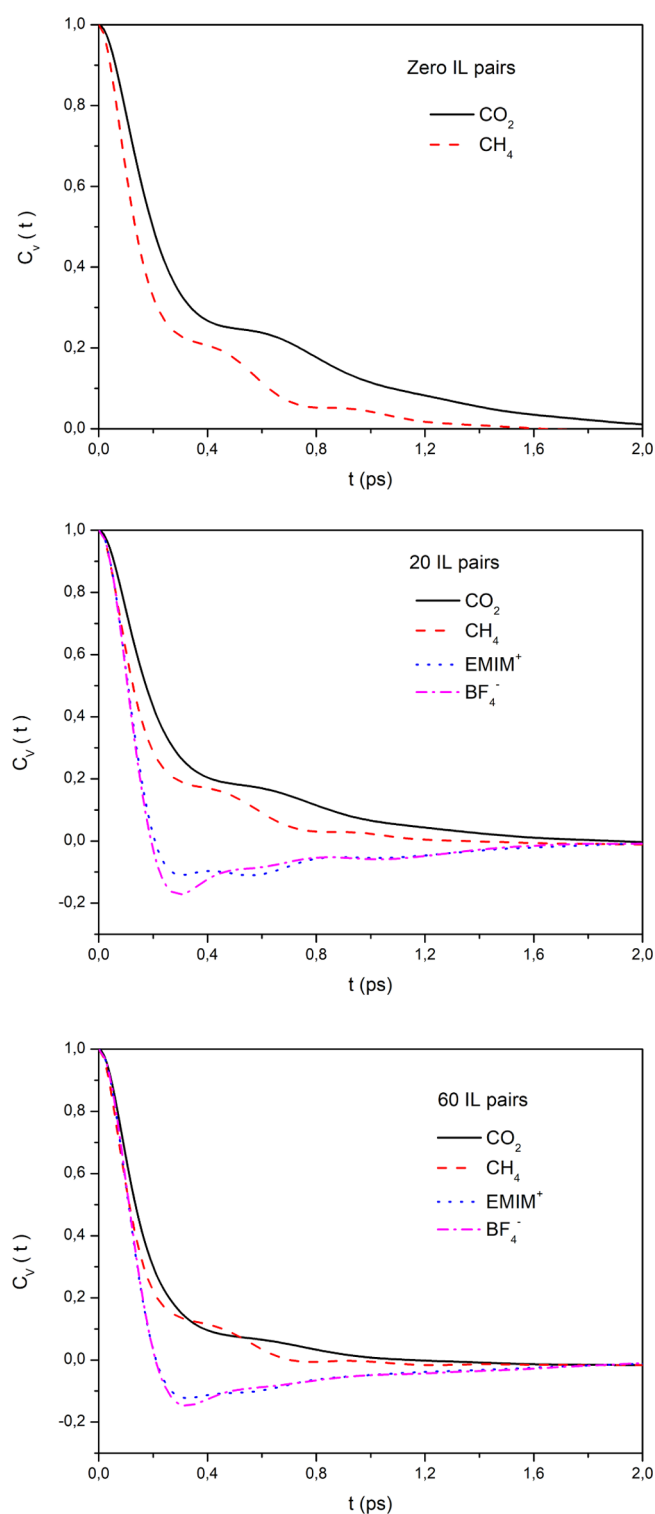


Figure 7. Calculated COM velocity tcf of all the adsorbed species at 1 bar, presented for the investigated PNN and PNN-IL composite materials.

induced by the denser packing inside the nanopores. The normalized COM velocity tcf of CO_2 and CH_4 do not exhibit negative values, indicating that their translational dynamics is less hindered in comparison with the IL ions. This is also reflected on the shape of their corresponding spectral densities, where small shoulders are observed around 50 and 70 cm^{-1} for CO_2 and CH_4 , respectively, in all the investigated cases. In the

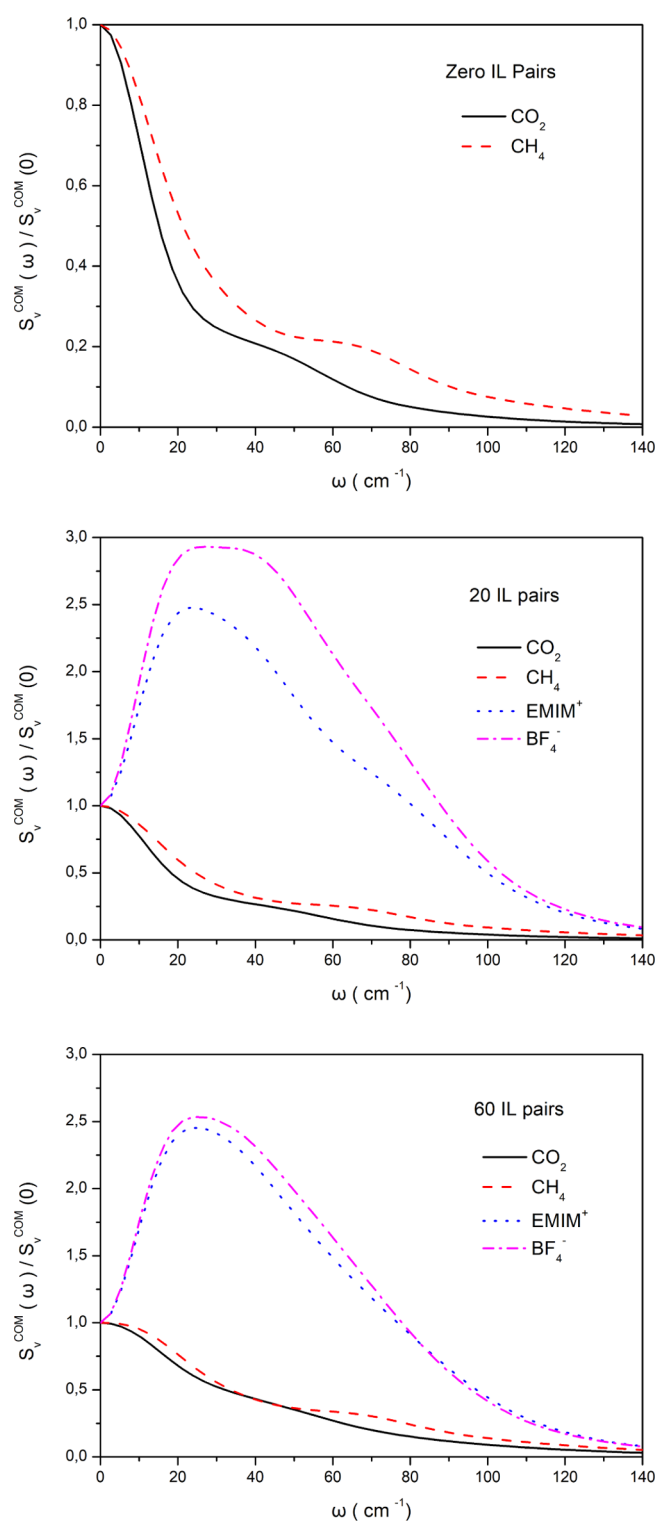


Figure 8. Calculated spectral densities of the COM velocity tcf of the adsorbed species at 1 bar, presented for the investigated PNN and PNN-IL composite materials.

case of the IL ions, their spectral densities exhibit a clear broad peak located around 30 cm^{-1} .

Previous studies⁷⁶ have pointed out that under dense packing conditions, the molecular shape can significantly affect the translational dynamics of the molecules. Depending on the molecular symmetry, some specific translational and rotational modes are thus more pronounced. In the case of linear

molecules, previous studies have proposed a specific methodology to analyze the translational dynamics in terms of the dynamics of the COM velocity components parallel and perpendicular to the molecular axis. In the present study, taking as reference the molecular axis of the CO₂ molecules at the initial time frame $t = 0$, the dynamics of these parallel and perpendicular components were studied, using the formalism proposed by Tildesley and Madden¹⁰¹

$$C_v^{\parallel}(t) = \frac{\langle (\vec{v}_i(0) \cdot \vec{u}_i(0)) \cdot (\vec{v}_i(t) \cdot \vec{u}_i(0)) \rangle}{\langle (\vec{v}_i(0) \cdot \vec{u}_i(0))^2 \rangle} \quad (5)$$

$$C_v^{\perp}(t) = \frac{\langle (\vec{v}_i(0) \times \vec{u}_i(0)) \cdot (\vec{v}_i(t) \times \vec{u}_i(0)) \rangle}{\langle (\vec{v}_i(0) \times \vec{u}_i(0))^2 \rangle} \quad (6)$$

In these equations, $\vec{u}_i$ is a unit vector along the C–O bond vector of CO₂. Note that other formalisms,^{102,103} where the instantaneous projection of the COM velocity on the molecular axis of a linear molecule at each time t is correlated, can also be applied to study these dynamics. The resulting similarities and differences in the dynamics of model rod-shaped molecules when using these formalisms have been systematically explored in recent studies.¹⁰³ The calculated $C_v^{\parallel}(t)$ and $C_v^{\perp}(t)$ tcfs for the CO₂ molecules are presented in Figure 9 for the PNN and PNN-IL composite materials with different IL concentrations, depicted together with the overall COM velocity tcfs of CO₂. From these figures, it can clearly be observed that the shape and decay of the overall COM velocity tcf $C_v(t)$ are mainly controlled by the decay of $C_v^{\perp}(t)$. This motion is more hindered in comparison with the motion along the CO₂ axis, as it can also be observed by the shape of the $C_v^{\parallel}(t)$ and $C_v^{\perp}(t)$ tcfs. The decay of $C_v^{\parallel}(t)$, which does not exhibit local minima, can be described in terms of a single exponential decay function and is slower than the decay of $C_v^{\perp}(t)$, due to the fewer hindrance to diffusion along the initial molecular axis direction than in the plane perpendicular to it. The hindrance in the diffusion along this plane becomes more pronounced as the IL concentration increases, due to the denser packing effects.

Apart from the Green–Kubo formalism used to study the translational dynamics of the confined species, their self-diffusion coefficient can be studied by calculating the molecular mean-square displacement and employing the well-known Einstein relation

$$D = \frac{1}{6} \lim_{t \rightarrow \infty} \frac{\partial}{\partial t} \langle \Delta \vec{r}_i(t)^2 \rangle = \frac{1}{6} \lim_{t \rightarrow \infty} \frac{\partial}{\partial t} \langle |\vec{r}_i(t) - \vec{r}_i(0)|^2 \rangle \quad (7)$$

The Einstein relation provides a general description of the molecular self-diffusion in condensed phases. Particular cases treating anomalous and sub-diffusion phenomena, as well as finite-size effects have also been presented in the literature.^{104–107} The calculated mean-square displacements $\text{MSD} = \langle \Delta \vec{r}_i(t)^2 \rangle$ of the CO₂ and CH₄ molecules and the Emim⁺ and BF₄[−] ions are presented for all the simulated systems in Figure 10. The dependence of the calculated self-diffusion coefficient, calculated from the slope of the mean square displacement during the time-scale 5–20 ns, of the adsorbed CO₂ and CH₄ molecules on the IL concentration is also presented in Figure 11. From these figures, we may observe that the IL concentration is crucial in controlling the diffusion of the adsorbed CO₂ and CH₄ molecules. As the IL concentrations increase, the corresponding MSD and self-diffusion coefficients of the adsorbed CO₂ and CH₄ molecules decrease. The

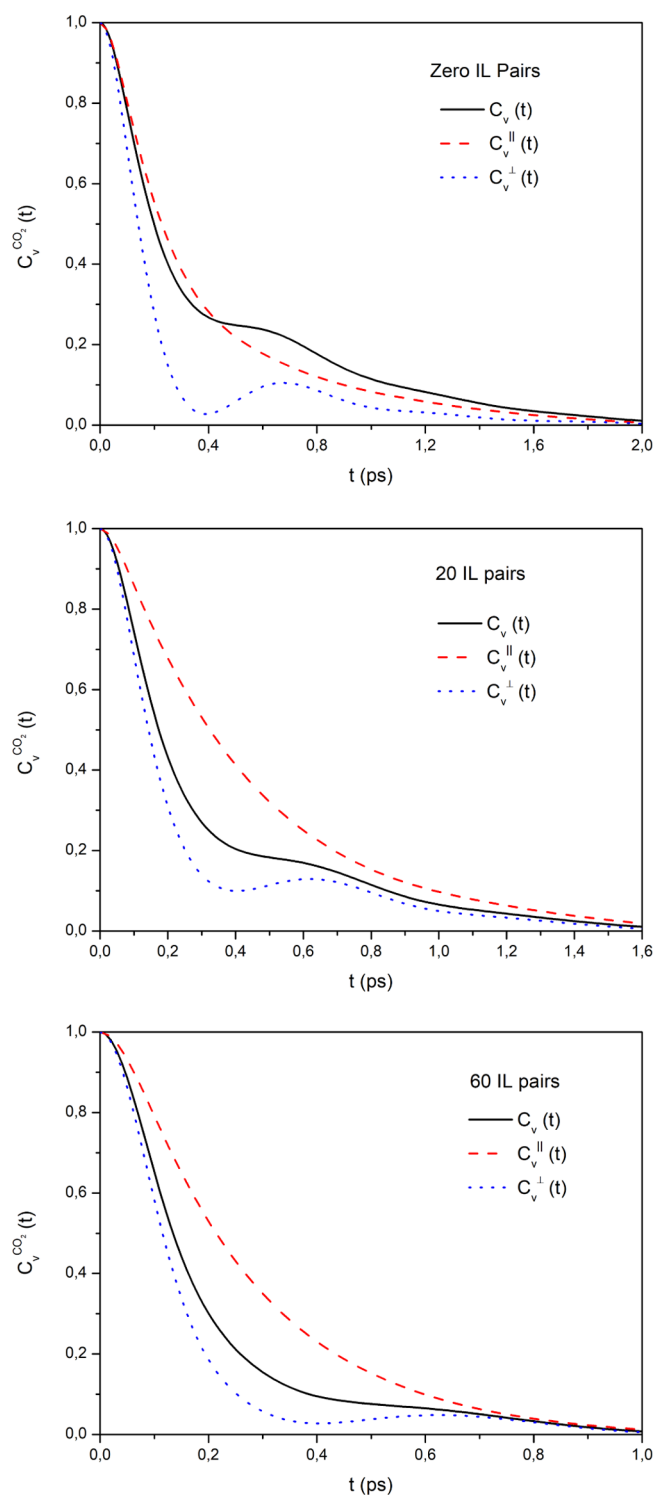


Figure 9. Calculated velocity tcfs $C_v^{\parallel}(t)$ and $C_v^{\perp}(t)$ for the adsorbed CO₂ molecules at 1 bar in the investigated PNN and PNN-IL composite materials, depicted together with the overall COM velocity tcfs.

calculated self-diffusion coefficients of CO₂ adsorbed in the non-IL-doped PNN, as well as in the IL-doped one (containing 20 and 60 added IL ion pairs) are 18.67×10^{-9} , 12.90×10^{-9} , and 5.36×10^{-9} m²/s, respectively. The corresponding values for CH₄ are 34.58×10^{-9} , 24.05×10^{-9} , and 11.08×10^{-9} m²/s. The corresponding self-diffusion ratio $D_{\text{CH}_4}/D_{\text{CO}_2}$ for these IL concentrations is therefore about 1.85, 1.86, and 2.07,

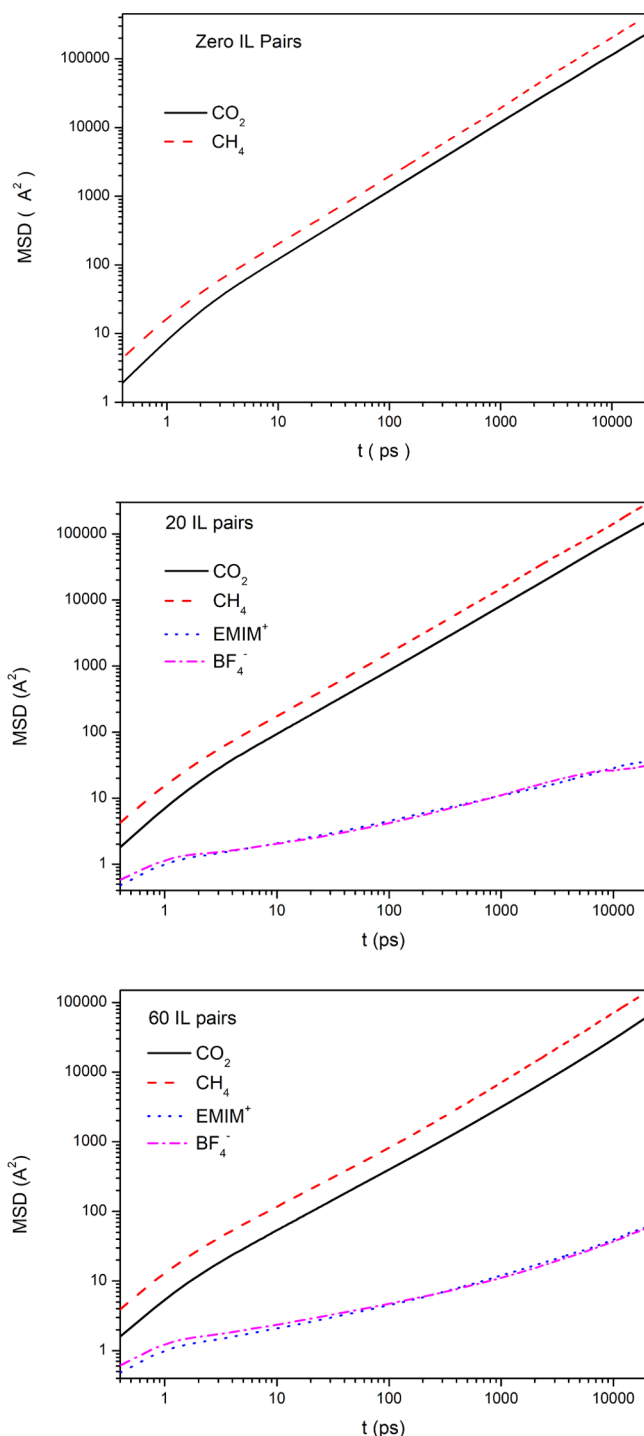


Figure 10. Calculated mean-square displacements $MSD = \langle \Delta \vec{r}_i(t)^2 \rangle$ of the adsorbed CO_2 and CH_4 molecules at 1 bar and the $Emim^+$ and BF_4^- ions, presented for the investigated PNN and PNN-IL composite materials.

respectively. This finding indicates that a satisfactory kinetic separation is also achieved for the PNN-IL composite materials at the investigated IL concentrations, similar to the one obtained for the non-IL-doped PNN. Therefore, for the investigated IL concentrations, the increase of the thermodynamic separation selectivity upon the IL loading is the main factor controlling the overall selectivity, which as mentioned in previous works⁶¹ depends on both the thermodynamic and the diffusion

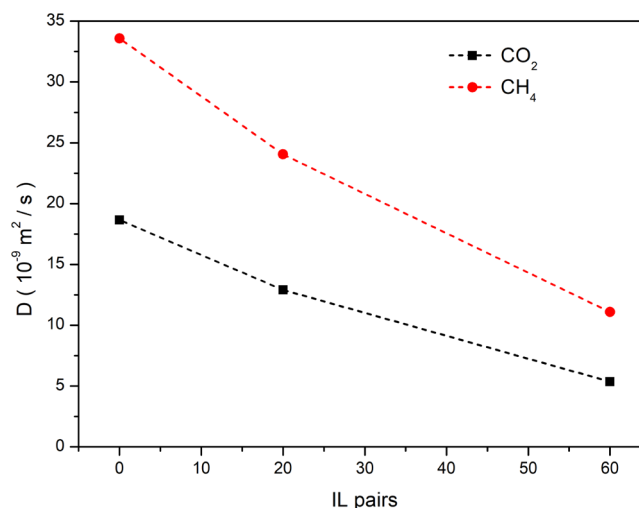


Figure 11. Calculated self-diffusion coefficients of the adsorbed CO_2 and CH_4 molecules at 1 bar in the investigated PNN and PNN-IL composite materials.

selectivity. It is also to be noted that at the investigated IL concentrations, the mean-square displacements of the $Emim^+$ and BF_4^- ions are lower by about 2–3 orders of magnitude in comparison with the corresponding ones for the CH_4 and CO_2 molecules. The mean-square displacement of the $Emim^+$ and BF_4^- ions is also very similar, indicating that the IL ions are moving inside the nanopores as ion pairs.

To further investigate the molecular shape effects on the translational dynamics of CO_2 , the displacement vector of CO_2 , $\Delta \vec{r}_i(t) = \vec{r}_i(t) - \vec{r}_i(0)$ has been analyzed in two components, one parallel to the molecular axis of the CO_2 molecules at the initial time frame $t = 0$, $\Delta \vec{r}_i^{\parallel}(t)$ and another component lying on the plane which is perpendicular to the same reference molecular axis, $\Delta \vec{r}_i^{\perp}(t)$. Since $\Delta \vec{r}_i^{\parallel}(t) \perp \Delta \vec{r}_i^{\perp}(t)$ and $\Delta \vec{r}_i(t) = \Delta \vec{r}_i^{\parallel}(t) + \Delta \vec{r}_i^{\perp}(t)$, then

$$MSD^{tot} = \langle \Delta \vec{r}_i(t)^2 \rangle = \langle \Delta \vec{r}_i^{\parallel}(t)^2 \rangle + \langle \Delta \vec{r}_i^{\perp}(t)^2 \rangle = MSD^{\parallel} + MSD^{\perp} \quad (8)$$

The components $MSD^{\parallel}$ and $MSD^{\perp}$ decompose the three-dimensional diffusion process, described by the overall MSD^{tot} , in two components corresponding to a mono- and a bi-dimensional diffusion process, respectively. In this respect, the corresponding time-dependent diffusion coefficients can be defined as follows

$$D^{\parallel}(t) = \frac{1}{2} \cdot \frac{\partial}{\partial t} \langle \Delta \vec{r}_i^{\parallel}(t)^2 \rangle, \quad D^{\perp}(t) = \frac{1}{4} \cdot \frac{\partial}{\partial t} \langle \Delta \vec{r}_i^{\perp}(t)^2 \rangle \quad (9)$$

At the long-time scale limit and if the dynamics can be described by the Einstein equation, then the overall self-diffusion D and the self-diffusion coefficients $D^{\parallel}$ and $D^{\perp}$ can be expressed as

$$D = \frac{D^{\parallel} + 2D^{\perp}}{3} \quad (10)$$

The calculated components $MSD^{\parallel}$ and $MSD^{\perp}$, together with the overall MSD^{tot} of CO_2 are presented for PNN and the PNN-IL composite materials with the investigated IL concentrations in Figure 12. The time dependence of the self-diffusion

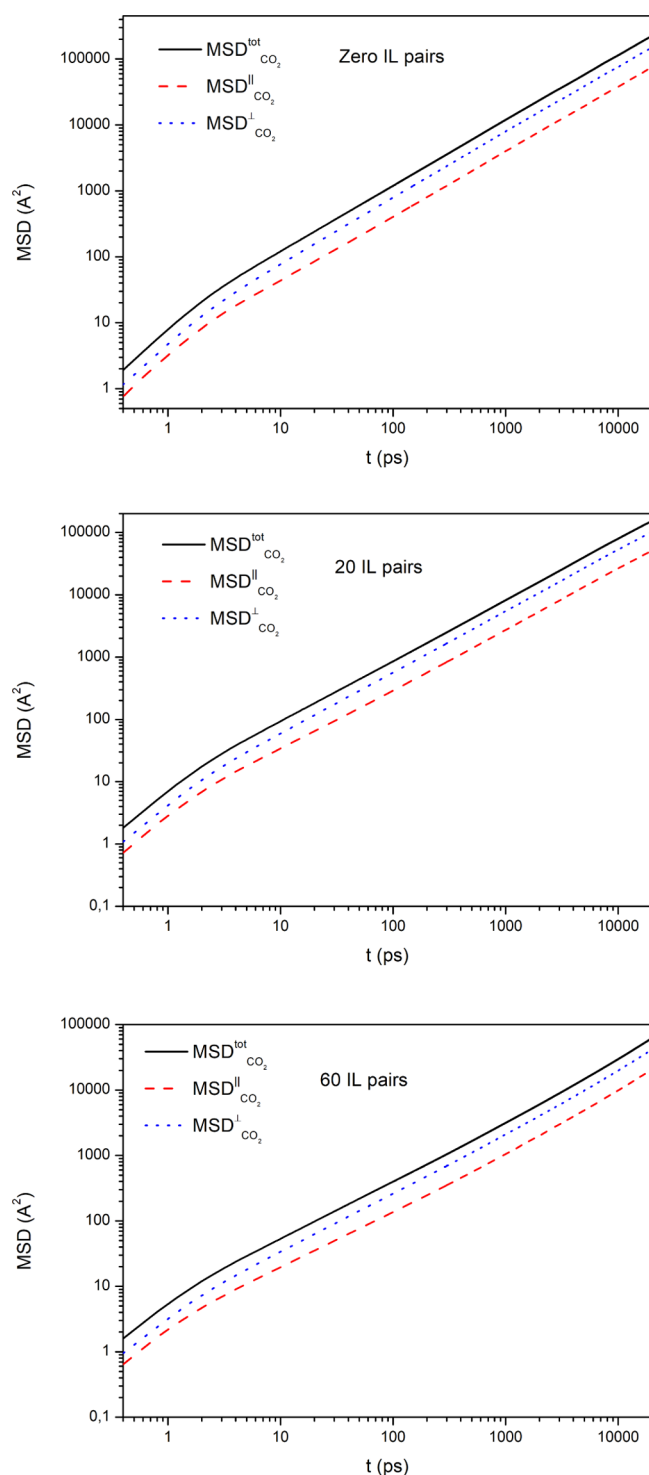


Figure 12. Calculated components $\text{MSD}^{\parallel}$ and $\text{MSD}^{\perp}$ of the adsorbed CO_2 molecules at 1 bar, plotted together with the overall MSD^{tot} of CO_2 , presented for the investigated PNN and the PNN-IL composite materials.

coefficients $D^{\parallel}$ and $D^{\perp}$ is also presented in Figure 13. From the results obtained, it can be concluded that in the long-time scale limit $\text{MSD}^{\perp} \approx 2 \cdot \text{MSD}^{\parallel}$ and $D^{\parallel} \approx D^{\perp} \approx D$, signifying that at these time scales the translational dynamics exhibit a diffusive behavior. However, at short-time scales up to about 100 ps, where local cage effect dynamics have a strong effect upon the single-molecule translational dynamics, significant deviations

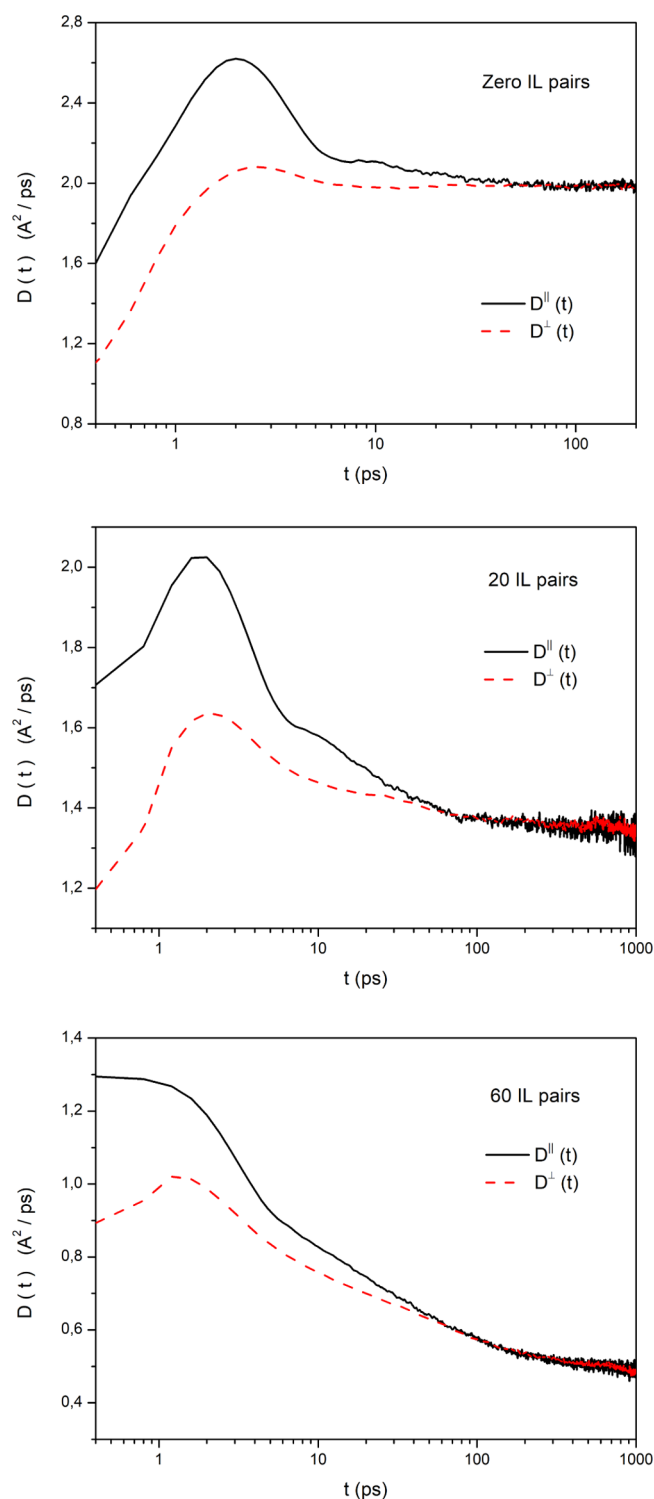


Figure 13. Time dependence of the self-diffusion coefficients $D^{\parallel}$ and $D^{\perp}$ for the adsorbed CO_2 molecules at 1 bar, presented for the investigated PNN and PNN-IL composite materials.

from the abovementioned relations are observed and the self-diffusion coefficient $D^{\parallel}(t)$ is higher than $D^{\perp}(t)$. This finding indicates that the diffusion along the direction of the molecular axis vector at $t = 0$ is significantly less hindered. Interestingly, the time scale where $D^{\parallel}$ becomes equal with $D^{\perp}$ is longer in the cases where the IL is loaded in the PNN. On the other hand, when the IL is not loaded in the PNN, $D^{\parallel}(t)$ becomes equal with $D^{\perp}(t)$ in a shorter time scale. This conclusion is in agreement with our

previous findings regarding the behavior of the corresponding velocity tcfs. All these findings further indicate that the linear shape of the CO₂ molecules plays an important role in the translational dynamics in highly packed environments.

The heterogeneous translational dynamics of the adsorbed species is also clearly reflected on the well-known non-Gaussian parameter $\alpha_2(t)$, which in the case of a d-dimensional diffusion process, is defined as¹⁰⁸

$$\alpha_2(t) = \frac{\langle \Delta \vec{r}_i(t)^4 \rangle}{\left(1 + \frac{2}{d}\right) \cdot \langle \Delta \vec{r}_i(t)^2 \rangle^2} - 1 \quad (11)$$

In the case of three-dimensional diffusion, $\alpha_2(t)$ has the value

$$\alpha_2(t) = \frac{3}{5} \cdot \frac{\langle \Delta \vec{r}_i(t)^4 \rangle}{\langle \Delta \vec{r}_i(t)^2 \rangle^2} - 1 \quad (12)$$

Following eq 11, two non-Gaussian parameters $\alpha_2^{\parallel}(t)$ and $\alpha_2^{\perp}(t)$ can be defined, in order to study the effect of local cage dynamics on these different translational modes

$$\begin{aligned} \alpha_2^{\parallel}(t) &= \frac{1}{3} \cdot \frac{\langle \Delta \vec{r}_i^{\parallel}(t)^4 \rangle}{\langle \Delta \vec{r}_i^{\parallel}(t)^2 \rangle^2} - 1, \\ \alpha_2^{\perp}(t) &= \frac{1}{2} \cdot \frac{\langle \Delta \vec{r}_i^{\perp}(t)^4 \rangle}{\langle \Delta \vec{r}_i^{\perp}(t)^2 \rangle^2} - 1 \end{aligned} \quad (13)$$

The calculated $\alpha_2(t)$ for both CO₂ and CH₄ molecules, as well as the $\alpha_2^{\parallel}(t)$ and $\alpha_2^{\perp}(t)$ for CO₂ are presented in Figure 14. From this figure, it can be observed that the position of the maxima in the cases of methane and CO₂ is located at around 4.5 and 7.5 ps, respectively. Upon the addition of the IL, the peak intensities of $\alpha_2(t)$, $\alpha_2^{\parallel}(t)$, and $\alpha_2^{\perp}(t)$ increase significantly, whereas their time decay also becomes slower. These findings signify the stronger local cage effects upon the translational dynamics when the IL added in the PNN. At short-time scales, $\alpha_2^{\perp}(t)$ also exhibits higher values than $\alpha_2^{\parallel}(t)$. This is an additional indication that the translational dynamics in the reference plane perpendicular to the molecular axis at $t = 0$ are more hindered, causing a higher deviation from a Gaussian behavior.

4. CONCLUSIONS

The effect of the addition of ILs in the nanopores of carbon-based materials on the separation of CO₂/CH₄ fluid mixtures was investigated using a combination of MC and MD simulation techniques. A three-dimensional carbon-nanotube network and the Emim⁺-BF₄⁻ IL have been used as prototype systems in the present treatment.

According to the results obtained, at pressures lower than 1 bar, the IL addition contributes to the increase of the CO₂ gravimetric uptake and the decrease of the CH₄ one. However, at higher pressures, the gravimetric uptake of both fluids is reduced, due to the decrease of the free pore volume when the IL concentration increases. The thermodynamic adsorption selectivity increases with the increase of the IL concentration for the whole investigated pressure range, and it becomes higher as the pressure increases, due to the significant decrease of the gravimetric methane uptake at high IL concentrations and pressures. A performance factor, describing the modification of the selectivity upon the addition of the IL ion pairs, was also introduced in the present study. The pressure dependence of this performance factor, as well as of the gravimetric uptake of CO₂ and the thermodynamic adsorption selectivity clearly indicate that there is an optimum pressure range, around 1 bar,

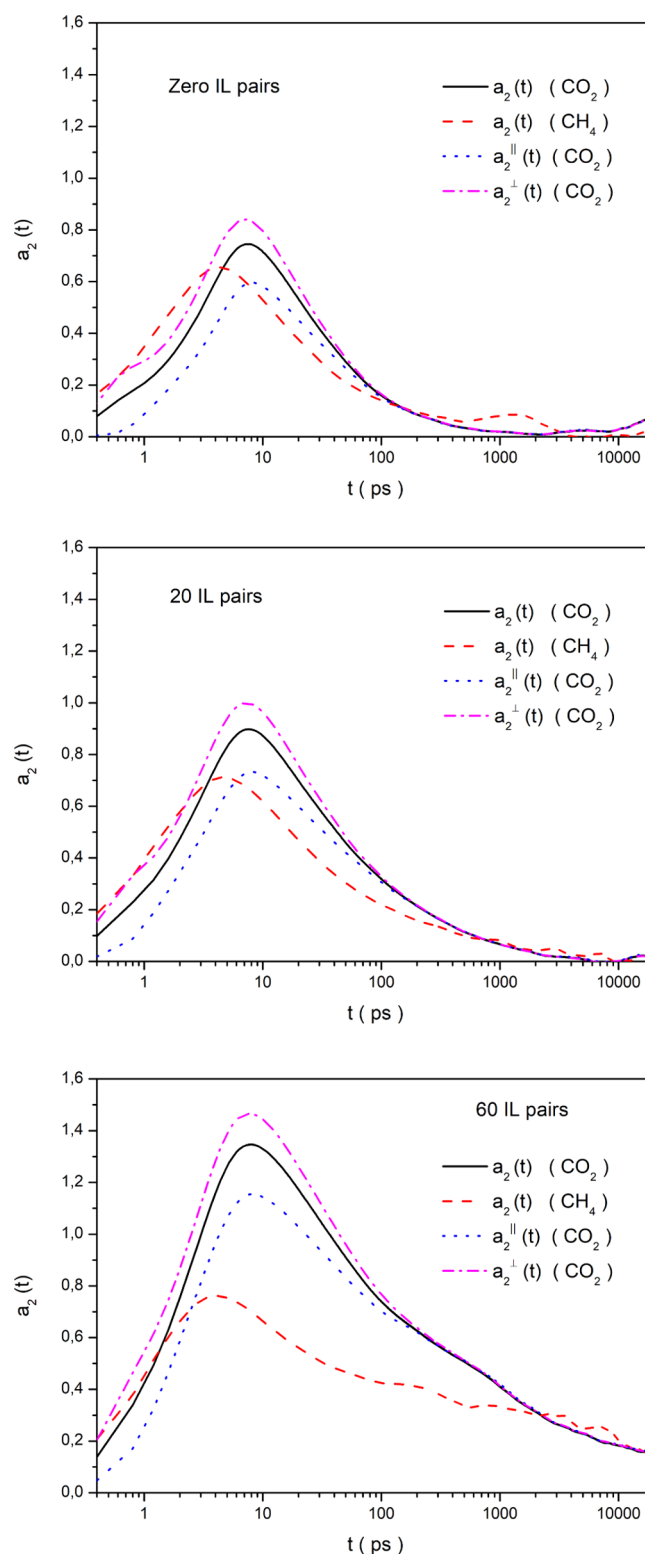


Figure 14. Time dependence of the non-Gaussian parameters $\alpha_2(t)$, $\alpha_2^{\parallel}(t)$, and $\alpha_2^{\perp}(t)$ for the confined CO₂ and CH₄ molecules at 1 bar, presented for the investigated PNN and PNN-IL composite materials.

simultaneously fulfilling the criteria for a quite satisfactory gravimetric uptake and selectivity.

The results obtained have revealed that the tuning of the IL concentration is very crucial in determining the mechanisms of the translational dynamics of the confined molecules at different time scales. The self-diffusion coefficients of confined CO₂ and

CH₄ molecules decrease significantly upon the IL addition, whereas the mean-square displacements of the Emim⁺ and BF₄[−] ions are lowered by about 2–3 orders of magnitude in comparison with the corresponding ones for the CO₂ and CH₄ molecules. The different behavior of the translational dynamics of CO₂ at short- and long-time scales is also reflected on the behavior of the $D^{\parallel}(t)$ and $D^{\perp}(t)$ self-diffusion coefficients, as well as on the time dependence of the non-Gaussian parameters $\alpha_2(t)$, $\alpha_2^{\parallel}(t)$, and $\alpha_2^{\perp}(t)$, further indicating that the local cage effects induced by the IL addition have a strong impact on the translational dynamics of the adsorbed molecules. The results obtained indicate that the molecular shape can be a crucial factor in determining the short- and long-time scale translational dynamics of the adsorbed molecules, revealing that the linear shape of the CO₂ molecules plays an important role on their translational dynamics under confinement. The findings and trends reported in the present treatment might serve as a springboard toward the rational design of IL carbon-based hybrid composite nanomaterials suitable as media for the separation of CO₂/CH₄ fluid mixtures. It should be noted, however, that this particular study deals with a specific carbon-based nanostructure and a specific prototype IL. Further work to investigate these phenomena for a wide variety of nanoporous materials and ILs is in progress.

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Notes

The authors declare no competing financial interest.

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